

Boxplot Analysis: Secondary Education Percentages

Seaborn Visualization Analysis & Summary Report

Visualization Overview

This report analyzes the boxplot distribution of **Secondary Education Percentage (ssc_p)** across different **Education Boards (ssc_b)**, split by **Gender (M vs F)**.

Summary Table of Metrics

Category (ssc_b)	Gender	Min (Whisker)	Lower Q1	Median Q2	Upper Q3	Max (Whisker)	Outliers
Others	Male (M)	~49%	~61%	~67%	~77%	~89.5%	None
Others	Female (F)	~45%	~65%	~74%	~78.5%	~87%	~41% (1 lower)
Central	Male (M)	~41%	~60%	~65%	~73%	~87%	None
Central	Female (F)	~43%	~60.5%	~67%	~75%	~88%	None

Detailed Comparative Analysis

- **Gender Comparison within "Others" Board:** Females score significantly higher than males with a median score of ~74% compared to ~67% for males. The female IQR is tighter (~65% to 78.5%) than male spread (~61% to 77%).
- **Gender Comparison within "Central" Board:** Females perform slightly better (Median ~67%) than males (Median ~65%). Spreads and whiskers are very comparable.
- **Board Comparison ("Others" vs "Central"):** Female students in the "Others" board outperform those in "Central" substantially (Median ~74% vs ~67%). Male performance across both boards remains closely aligned (~67% vs ~65%).